

Method

Protocol Preparation Time:

Total preparation time for nanopore long-read sequencing can vary depending on the molecule type. For DNA it ranges from 10 min (Rapid Sequencing Kit) – 3 hours 20 min with an overnight incubation period (Ultra-long DNA Sequencing Kit), and RNA can be between 2 hours 20 min (Direct RNA Sequencing Kit) - 3 hours 45 min + PCR time (cDNA-PCR Sequencing Kit).

Main types of sequencing:

1. Indirect
 - a. PCR amplification, loses modified basecalling ability
2. Direct
 - a. No PCR amplification, retains modified basecalling

What can you sequence?

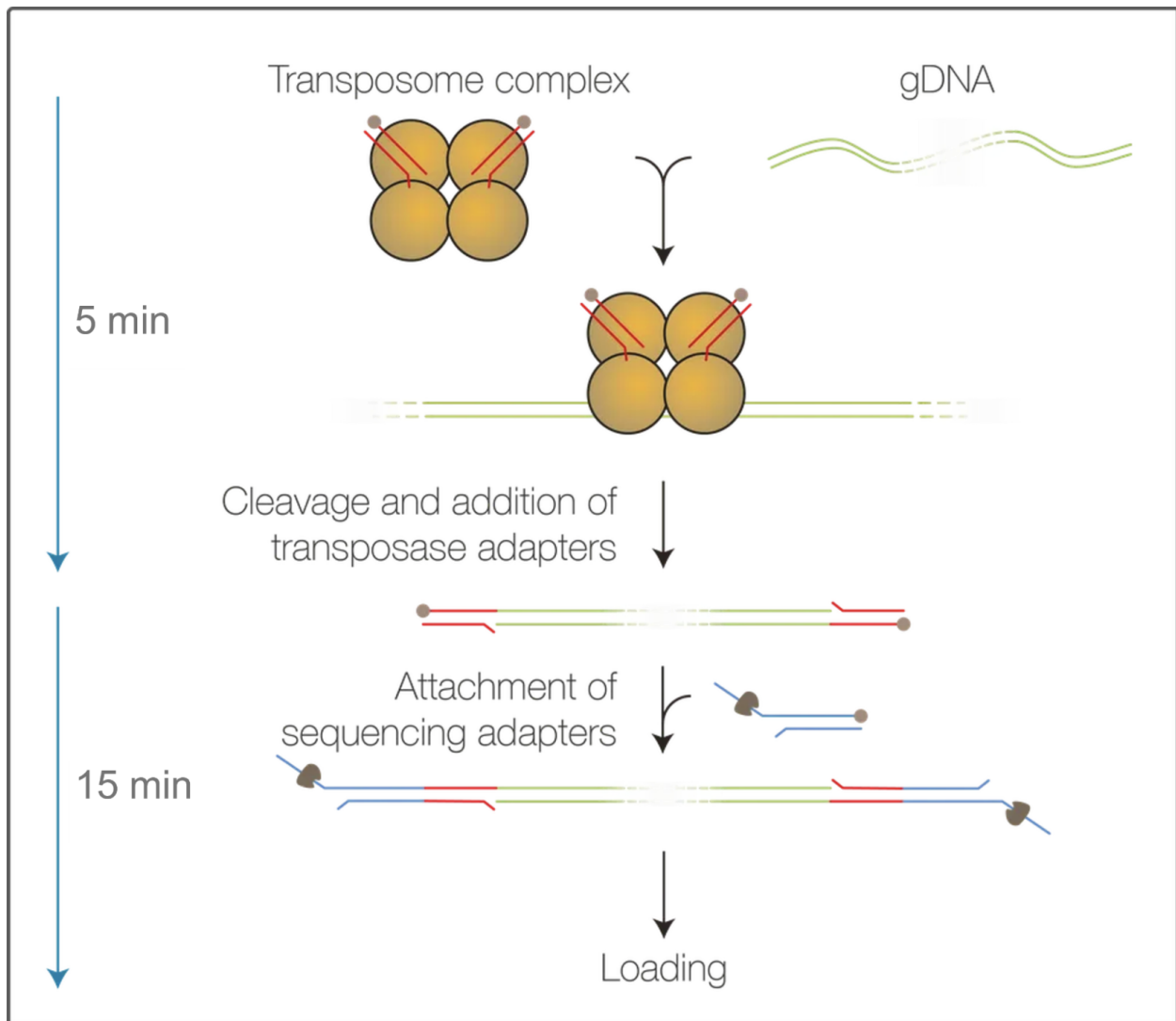
Just about any sized DNA or RNA molecule (~20 bp – 4 Mb)!

Any organism's nucleic material can be sequenced, as basecalling can be performed with alignment to a reference genome, or it can basecall without alignment, allowing for de novo assembly.

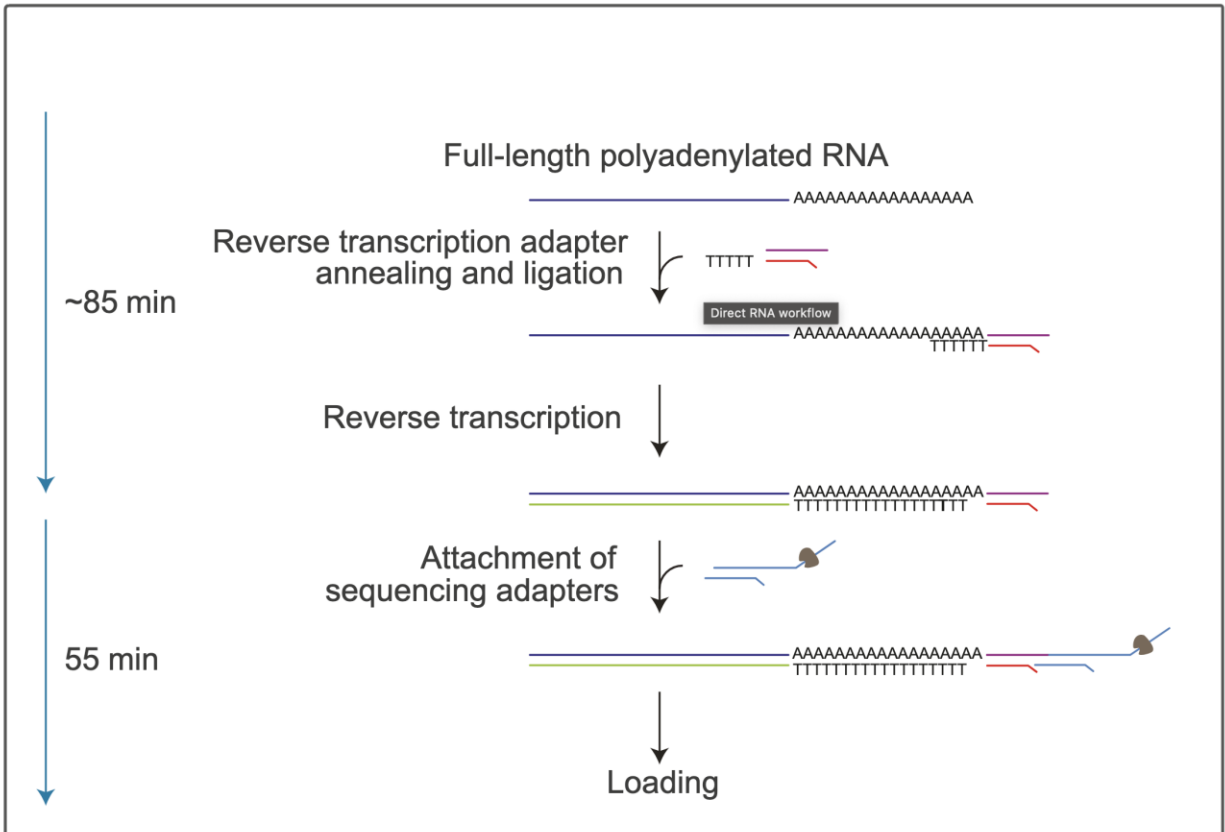
Nanopore Sequencing Mechanism

Each kit has reagents specific to sequencing needs, but they rely on the addition of sequencing adapters.

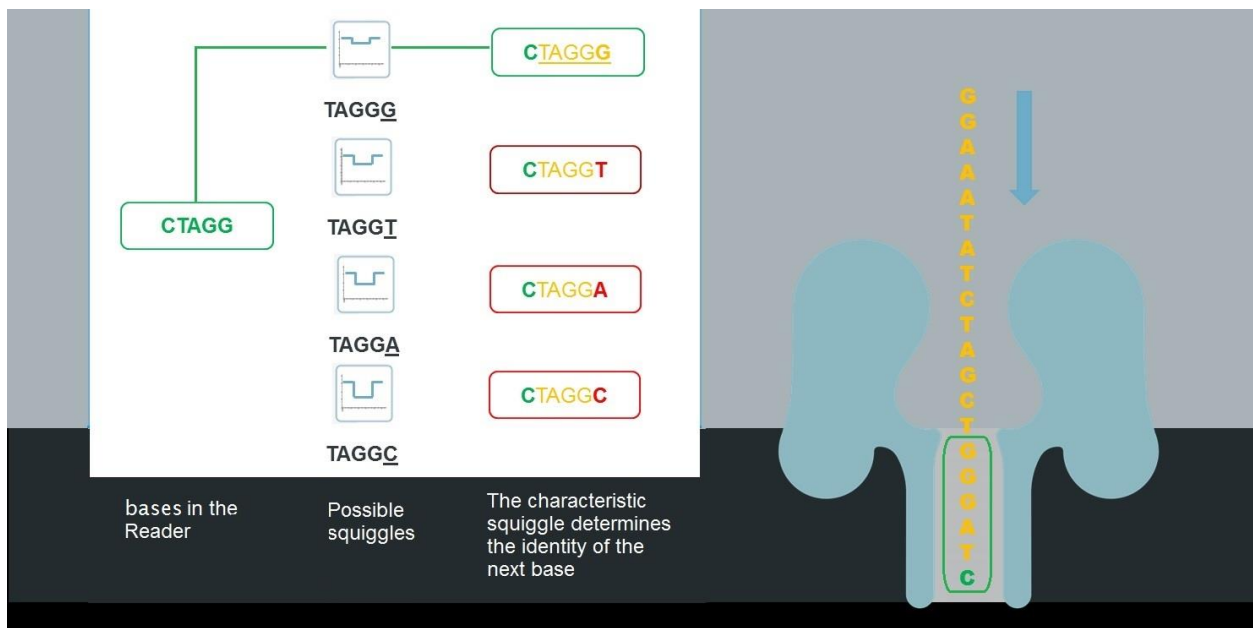
Rapid Sequencing Kit:



Direct RNA Sequencing Kit:



The Flow Cell (where the magic happens):



References: Oxford Nanopore Technologies

(<https://nanoporetech.com/resources/document-repository>)

Why Nanopore Sequencing

Nanopore Long-Read Sequencing	
Advantages	Disadvantages
Ultra-long read capability ¹ with tunable selectivity ²	Lowered basecall accuracy (91-98.7%) ⁶
Versatile: DNA/RNA/modifications (soon polypeptides?) ³	Rapid technological advancements that require continuous updates ⁴
Reusable flow cell with portable/user friendly device ⁴	Large raw data file sizes (total > 1.0 TB) ⁷
Greater yield return with faster/simpler library preparation ^{4, 5}	Flow cell pore availability variable ⁶

References:

1. Jain et al., *Nat. Biotechnol.*, 2018 (<https://doi.org/10.1038/nbt.4060>)
2. Loose et al., *Nat. Methods*, 2016 (<https://doi.org/10.1038/nmeth.3930>)
3. Akesen et al., *US Patent #: US 11,970,738 B2*, 2024 (<https://patentimages.storage.googleapis.com/99/36/59/5146a1db78a52c/US11970738.pdf>)
4. ONT Documents (<https://nanoporetech.com/resources/document-repository?filter.content-type=documentType%3EMethodology+documentation&filter.content-type=documentType%3EWorkflow>)
5. Klier, *Master's Projects*, 2022 (<https://doi.org/10.31979/etd.n6cs-f4jw>)
6. Santos et al., *Int. J. Mol. Sci.*, 2025 (<https://doi.org/10.3390/ijms26104492>)
7. Jayasooriya et al., *Genome Res.*, 2025 (<https://doi.org/10.1101/gr.280090.124>)